

A Six-State Boundary Lemma for Two Blanuša Six-Poles

Provisional computational note—not a resolution of FIVECDC

Atharva Vaidya

Draft of 29 July 2026

Abstract

We isolate a finite positive boundary relation for two ordered six-poles obtained by deleting two vertices from the two Blanuša snarks. Edges are labelled by the ten weight-two vectors of $\mathbb{F}_2^5$, the standard flow encoding of five labelled even subgraphs covering every edge twice. Six particular three-edge boundary states have the property that every ordered concatenation of two such states extends through either six-pole, for every port order. This gives a constructive composition lemma. A solver-free proof table lists all $2 \cdot 36^2 = 2592$ internal extensions, and a separately implemented checker verifies the graph encodings and every parity equation. As one application, a canonical census verifies the hypothesis for 9,278 independent junction-set orbits in cyclically 4-edge-connected simple cubic macros of order at most 14. This is a restricted local and finite result; it neither proves nor disproves the Five-Cycle Double Cover Conjecture.

Status warning. This draft has not been peer reviewed or independently checked by a human. A bounded literature audit found no exact antecedent for the stated local relation, but that audit does not establish novelty. Section 9 discloses the use of generative AI.

1 Five labelled even subgraphs as a duad flow

All graphs in the theorem below are finite. For a graph G , an *even subgraph* means an edge subset having even degree at every vertex; it need not be connected and may be empty. This is the convention under which a labelled five-cycle double cover is a tuple $(C_0, \dots, C_4)$ of even subgraphs such that every edge belongs to exactly two members of the tuple.

Let

$$D = \{x \in \mathbb{F}_2^5 : \text{wt}(x) = 2\}.$$

We identify a vector supported on $\{i, j\}$ with the duad ij .

Proposition 1 (standard duad formulation). *A graph G has five labelled even subgraphs covering every edge exactly twice if and only if there is a map $\lambda : E(G) \rightarrow D$ such that*

$$\bigoplus_{e \ni v} \lambda(e) = 0 \quad \text{for every } v \in V(G),$$

where a loop is counted twice in the incidence sum.

Proof. Given $(C_0, \dots, C_4)$, let the i th coordinate of $\lambda(e)$ indicate whether $e \in C_i$. Exact double coverage gives $\text{wt}(\lambda(e)) = 2$, and even degree in every C_i gives the displayed xor equation coordinate by coordinate. Conversely, take C_i to be the edges whose labels have i th coordinate one. The xor equations make every C_i even and the weight-two condition gives exact double coverage. $\square$

Proposition 1 is background, not a novelty claim. Equivalent weight-two flow formulations occur in the FiveCDC literature; see, in particular, Hušek and Šámal [6].

2 Ordered six-poles and their boundary relations

An ordered six-pole X consists here of a cubic graph fragment with six ordered semiedges. Its internal vertices have degree three after semiedges are counted. A word $w = (w_1, \dots, w_6) \in D^6$ is *extendable through* X if the internal edges can be labelled from D so that the xor of incident labels is zero at every internal vertex. Let

$$R_X = \{w \in D^6 : w \text{ is extendable through } X\}.$$

The two fragments are fixed by the following machine-independent data. The graph6 strings encode the source graphs; vertices are numbered from zero. After deletion, the verifier canonically relabels the remaining vertices and recovers the displayed ordered port vertices.

atom	source graph6	deleted source vertices	ordered ports after relabelling
A	Q??CA?_CCOW_Q_M?AD@A_@K?F??	4, 9	(8, 10, 11, 3, 13, 14)
B	Q???C@?GCo0oDO[?CcAO_?k?J??	13, 16	(1, 2, 3, 6, 8, 9)

Each fragment has 16 internal vertices, 21 internal edges, and six semiedges.

3 The six path states

Fix the coordinate 0 and put $Y = \{1, 2, 3, 4\}$. If $\{a, b\} \subset Y$ and $\{c, d\} = Y \setminus \{a, b\}$, define the unordered three-duad state

$$P_{ab} = \{0a, 0b, cd\}.$$

Thus the six states are

$$\begin{aligned} &\{01, 02, 34\}, \quad \{01, 03, 24\}, \quad \{01, 04, 23\}, \\ &\{02, 03, 14\}, \quad \{02, 04, 13\}, \quad \{03, 04, 12\}. \end{aligned}$$

Let $\mathcal{P}_{\text{ord}} \subset D^3$ contain all orderings of these states. It has $6 \cdot 3! = 36$ members.

Lemma 2 (two-atom path universality). *For each $X \in \{A, B\}$,*

$$\mathcal{P}_{\text{ord}} \times \mathcal{P}_{\text{ord}} \subseteq R_X.$$

In particular, every one of the $36^2 = 1296$ ordered boundary words extends through each atom.

Finite proof certificate. The file `path-atom-witnesses.json` in the accompanying frozen artifact contains one row for each of the 1,296 boundary words and each atom. A row contains the boundary word followed by 21 internal duads in the listed internal-edge order. For every row, direct inspection consists only of:

1. checking that each boundary and internal label is a five-bit integer of Hamming weight two;
2. placing the six boundary labels on the displayed ordered ports; and
3. xoring the three incident labels at each of the 16 internal vertices.

All resulting vertex xors are zero. Hence each row is an explicit extension. The accompanying solver-free program `verify_frontier.py` performs exactly these checks after reconstructing both source graphs from their graph6 strings. It imports no producer module and calls no SAT solver. $\square$

Remark 3. This is a checkable computer-assisted proof: the table supplies witnesses, not an assertion that a search returned “satisfiable.” It is not yet a compact hand proof of all 2,592 rows. A symmetry reduction to a small set of conceptual cases remains desirable.

4 The composition lemma

Let M be a cubic macro graph whose vertices are partitioned into *junctions* and *connectors*. Suppose the connectors are paired. Replace each pair by either A or B , attaching the three incident macro edges of the first connector to the first port block and those of the second connector to the second block, in arbitrary orders. Junctions and macro edges remain.

Theorem 4 (six-state composition). *Suppose the macro edges have labels in D such that:*

1. *the three labels incident with every junction xor to zero; and*
2. *the unordered labels incident with every connector form one of the six states P_{ab} .*

Then every connector matching, every assignment of A or B to the pairs, every ordering of each three-port block, and either orientation of each pair produces an expanded graph with five labelled even subgraphs covering every edge exactly twice.

Proof. Consider one matched connector pair. Its ordered six boundary labels form an element of $\mathcal{P}_{\text{ord}} \times \mathcal{P}_{\text{ord}}$. By Lemma 2, choose the saved internal completion for the selected atom and port orders. Make this choice independently for every pair.

The two occurrences of every surviving macro edge retain the same label. The vertex xor is zero at each junction by hypothesis and at each atom vertex by the selected local completion. The result is therefore a D -valued flow on the expanded graph. Proposition 1 converts it to the required five even subgraphs. $\square$

5 A finite cyclically 4-edge-connected application

A connected cubic macro is called cyclically 4-edge-connected if no edge cut of size at most three has a cycle on both shores. For every even, nonempty, independent junction set, connector vertices are paired arbitrarily in the expansion above.

Theorem 5 (verified macro frontier). *For each cyclically 4-edge-connected simple cubic macro of order 6, 8, 10, 12, or 14, and each automorphism orbit of even nonempty independent junction sets, there is a macro-edge labelling satisfying the hypotheses of Theorem 4. Consequently every associated expansion has a five-cycle double cover, for all connector matchings, atom choices, and port orders.*

Reproducible finite census. The source graphs are the canonical output of `geng -cq -d3 -D3 n`. Shore enumeration applies the stated cyclic edge-connectivity definition, full automorphism groups act on the junction sets, and the artifact stores a positive macro-edge witness for every representative. The counts are:

n	6	8	10	12	14	total
connected cubic sources	2	5	19	85	509	620
cyclically 4-connected macros	1	2	5	18	84	110
junction-set orbits	1	5	32	445	8795	9278.

For every saved witness, direct xor checks establish the two hypotheses of Theorem 4. The independent verifier rebuilds the census, automorphism orbits, and all witness checks. $\square$

6 Scope and limitations

Theorem 5 is not a reduction of general FiveCDC to macros of order at most 14. No induction showing that every cyclically 4-edge-connected cubic macro has a six-state labelling is known. Indeed, a diagnostic enlargement to all 3-edge-connected macros through order 14 has one six-state failure on a macro with a nontrivial cyclic 3-edge cut; adding one compatible star state repairs the diagnostic census. That negative diagnostic is not a FiveCDC counterexample and is not used in any theorem above.

No target graph was found for which the direct five-coordinate SAT/XOR formula is unsatisfiable. Therefore this note provides neither an UNSAT certificate nor a disproof of FiveCDC. It also does not address the orientable FiveCDC conjecture.

7 Relation to prior work and novelty status

Multipoles, their semiedge states, and their use in snark reductions are standard [9, 1]. Hušek and Šámal [4] give the closest general framework: ordered multipoles, circuit-double-cover boundaries, multiplicity vectors, and linear gluing. Their boundaries remember path identities in circuit double covers with an unrestricted number of circuits. Our boundary instead fixes five named even subgraphs and assigns each semiedge the two names containing it.

Liu, Hao, Luo, and Zhang [7] develop a recent FiveCDC superposition method using nowhere-zero 4-flows and Catlin reduction, and apply it to several snark families. This is close prior art for the global composition strategy. We found no ordered weight-two $\mathbb{F}_2^5$ six-pole state relation or either explicit Blanuša atom table in the inspected material.

Normal and Petersen edge colorings provide another related five-color language [8, 10], but their edge labels and local normality conditions are not the weight-two $\mathbb{F}_2^5$ flow condition used here.

Our bounded search of primary sources found no publication containing Lemma 2 for these two ordered Blanuša fragments. The honest status is therefore *possible local novelty, not established novelty*. Old theses, unpublished tables, and differently named multipole computations remain material gaps in the audit.

8 Artifact and checking instructions

The frozen artifact directory is

`cyclic4-hybrid-universal-through14-20260729.`

Its SHA256SUMS file has SHA-256 digest

`2fd36f21f18796c15760321806152fe60f37d5ba40575a11c26dc76c9c31f3d7.`

From that directory, run

`python3 verify.py.`

The expected terminal lines include:

```
PASS SHA-256 ledger
PASS cyclically-4 hybrid path-core frontier through order 14
cyclic4_macros=110 junction_orbits=9278
macro_witnesses=9278 atom_completions=2592
```

```
all_matchings_all_atom_types_all_port_orders=true
target_unsat=0 target_resolution=false
PASS frozen cyclically-4 hybrid package
```

9 AI-use and review disclosure

The human researcher selected the conjecture, set the research and publication direction, and requested explicit certification and honest disclosure. Collaborating OpenAI Codex agents wrote essentially all search programs and checker code, generated the witness tables, conducted the bounded literature search, and drafted this manuscript. The phrase “independent verifier” means a separately implemented code path; it does not mean an independently authored human verification. As of the date on the title page, no human expert has line-by-line verified the proof, reproduced the computation from a clean checkout, or assessed the novelty claim. These facts should remain in any public version, updated only to record reviews that actually occurred.

References

- [1] M. A. Fiol and J. Vilaltella, Some results on the structure of multipoles in the study of snarks, *Electronic Journal of Combinatorics* 22(1) (2015), P1.45. <https://doi.org/10.37236/3629>
- [2] A. Hoffmann-Ostenhof, A note on 5-cycle double covers, *Graphs and Combinatorics* 29 (2013), 977–979. <https://arxiv.org/abs/1209.0096>
- [3] A. Huck and M. Kochol, Five cycle double covers of some cubic graphs, *Journal of Combinatorial Theory, Series B* 64 (1995), 119–125.
- [4] R. Hušek and R. Šámal, Counting circuit double covers, *Journal of Graph Theory* 108 (2025), 374–395. <https://arxiv.org/abs/2303.10615>
- [5] R. Hušek and R. Šámal, Homomorphisms of Cayley graphs and Cycle Double Covers, *Electronic Journal of Combinatorics* 27(2) (2020), P2.2. <https://arxiv.org/abs/1901.03112>
- [6] R. Hušek and R. Šámal, Exponentially Many Circuit Double Covers, arXiv:2607.24724 (2026). <https://arxiv.org/abs/2607.24724>
- [7] S. Liu, R.-X. Hao, R. Luo, and C.-Q. Zhang, 5-Cycle Double Covers, 4-Flows, and Catlin Reduction, *SIAM Journal on Discrete Mathematics* 37 (2023), 253–267. <https://doi.org/10.1137/22M1472425>
- [8] G. Mazzuoccolo and V. Mkrtychyan, Normal 6-edge-colorings of some bridgeless cubic graphs, *Discrete Applied Mathematics* (2020). <https://arxiv.org/abs/1903.06043>
- [9] R. Nedela and M. Škoviera, Decompositions and reductions of snarks, *Journal of Graph Theory* 22 (1996), 253–279. [https://doi.org/10.1002/\(SICI\)1097-0118\(199607\)22:3%3C253::AID-JGT6%3E3.0.CO;2-L](https://doi.org/10.1002/(SICI)1097-0118(199607)22:3%3C253::AID-JGT6%3E3.0.CO;2-L)
- [10] F. Pirot, J.-S. Sereni, and R. Škrekovski, Variations on the Petersen colouring conjecture, *Electronic Journal of Combinatorics* 27(1) (2020), P1.8. <https://arxiv.org/abs/1905.07913>